

AI-Assisted Learning and the Reversal of Long-Run Skill Degradation

Topic presentation — Track A

wbgradost

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<https://github.com/wbgradost/ai-project>

Can AI assistance raise long-run skill when it makes the **remaining human effort more effective for learning?**

Current production

AI substitutes for effort



Skill acquisition

AI may complement effort

Baseline: Aouad, Lykouris, and Zhong (2026)

- ▶ Proposition 3.5, p. 8: more AI shifts stationary skill downward.

One changed assumption:

$$\lambda(e) \longrightarrow \lambda(e, a) = \lambda_0 + \kappa e(1 + \eta a).$$

Target: an exact two-state condition for a *local* reversal, plus the threshold at which effort crowd-out regains dominance.

A myopic worker chooses effort for current utility:

$$\max_{e \geq 0} p(s + e + a) - \gamma e, \quad \gamma > 0.$$

- ▶ p : increasing, concave; s, e, a are perfect substitutes.
- ▶ Learning does not enter the worker's current objective.

Kuhn–Tucker conditions:

$$e^* \geq 0, \quad \gamma - p'(s + e^* + a) \geq 0, \quad e^*[\gamma - p'(s + e^* + a)] = 0.$$

2 · Optimal effort: one-for-one crowd-out

Let

$$x^* = \max \arg \max_{x \geq 0} \{p(x) - \gamma x\}.$$

Then Proposition 2.1 gives

$$e^*(s, a) = (x^* - s - a)^+.$$

If $s + a < x^*$

$$\frac{\partial e^*}{\partial a} = -1, \quad s + e^* + a = x^*.$$

If $s + a \geq x^*$

$$e^* = 0.$$

Baseline result: Aouad, Lykouris, and Zhong (2026), Proposition 2.1.

Continuous-time birth–death chain:

$$s_i \xrightarrow{\lambda(e^*(s_i, a))} s_{i+1}, \quad s_i \xrightarrow{\mu} s_{i-1}.$$

Local balance implies

$$\frac{\pi_{i+1}(a)}{\pi_i(a)} = \frac{\lambda(e^*(s_i, a))}{\mu}.$$

Higher $a \Rightarrow$ lower $e^* \Rightarrow$ lower upward rates $\Rightarrow$ more stationary mass on low skill.

Proposition 3.5 (baseline)

For $a_\ell < a_h$, the distribution under a_ℓ first-order stochastically dominates the distribution under a_h .

$$\boxed{\lambda(e, a) = \lambda_0 + \kappa e(1 + \eta a)}$$

$$\lambda_0 > 0, \quad \kappa > 0, \quad \eta \geq 0.$$

$$\lambda_e = \kappa(1 + \eta a) > 0,$$

$$\lambda_{ea} = \kappa\eta \geq 0.$$

- ▶ For fixed a : linear and weakly concave in effort.
- ▶ $\eta = 0$: affine special case of the original technology.
- ▶ Worker problem unchanged: effort remains $(x^* - s - a)^+$.

3 · Two states: stationary distribution

Let $s_L < s_H$ and define

$$\lambda_L(a) = \lambda(e^*(s_L, a), a).$$

$$s_L \xrightleftharpoons[\mu]{\lambda_L(a)} s_H$$

Stationary flow balance:

$$\pi_L \lambda_L = \pi_H \mu, \quad \pi_L + \pi_H = 1.$$

Therefore

$$\pi_H(a) = \frac{\lambda_L(a)}{\lambda_L(a) + \mu}.$$

The sign of $d\pi_H/da$ is exactly the sign of $d\lambda_L/da$.

3 · Competing effects of AI

Set $b = x^* - s_L > 0$. While $0 \leq a < b$,

$$e_L^*(a) = b - a$$

and

$$\lambda_L(a) = \lambda_0 + \kappa(b - a)(1 + \eta a).$$

Total effect:

$$\begin{aligned} \frac{d\lambda_L}{da} &= \underbrace{\kappa\eta e_L^*}_{\text{learning complementarity}} - \underbrace{\kappa(1 + \eta a)}_{\text{effort crowd-out}} \\ &= \kappa[\eta(b - 2a) - 1]. \end{aligned}$$

$$\frac{d\pi_H}{da} > 0 \quad \Longleftrightarrow \quad \eta(b - 2a) > 1.$$

3 · Candidate proposition: two thresholds

Weak complementarity

$$\eta b \leq 1$$

$\pi_H(a)$ is weakly decreasing: baseline degradation survives.

Strong complementarity

$$\eta b > 1$$

Initial reversal, ending at

$$a^\dagger = \frac{\eta b - 1}{2\eta}.$$

$$0 \xrightarrow{\pi_H \uparrow} a^\dagger \xrightarrow{\pi_H \downarrow} b \xrightarrow{\pi_H \text{ constant}}$$

At $a = b$, effort is zero; learning returns to λ_0 .

3 · Exact pairwise reversal condition

For two active-effort levels $0 \leq a_\ell < a_h < b$,

$$\lambda_L(a_h) - \lambda_L(a_\ell) = \kappa(a_h - a_\ell)[\eta(b - a_h - a_\ell) - 1].$$

Since $\pi_H = \lambda_L/(\lambda_L + \mu)$,

$$\boxed{\pi_H(a_h) > \pi_H(a_\ell) \iff \eta(b - a_h - a_\ell) > 1.}$$

Scope

This reverses Proposition 3.5 only for qualifying pairs of assistance levels. If $a_h \geq b > a_\ell$, the higher level eliminates effort and the reversal cannot hold.

4 · What the baseline appendices establish

- ▶ Appendix A.1 derives the stationary birth–death distribution.
- ▶ Appendix A.4 proves Proposition 3.5 using $\lambda = \lambda(e)$ and monotonicity in effort.
- ▶ Appendix D allows convex effort costs, but the learning rate still has no direct AI argument.
- ▶ The current arXiv record contains v1; the authors list an EC 2026 conference version.

Missing object: no AI–effort cross-partial in the transition technology, hence no closed-form stationary reversal condition.

Course Design in the Age of AI

- ▶ Myopic student chooses work or full delegation; teacher designs tasks.
- ▶ AI quality affects both the marginal product of effort and delegation.
- ▶ Theorem 4: higher quality raises learning for high-skill students but lowers it for low-skill students.

What this project adds to the Aouad et al. model

Keep its worker problem and stationary Markov structure; place complementarity directly in $\lambda(e, a)$; derive a closed-form stationary reversal condition and assistance threshold.

No claim that AI–learning complementarity itself is new.

Davies, arXiv:2607.18735v3, Theorem 4; Shen–Tamkin (2026); Wu et al. (2026).

1. **Proof:** all two-state boundary/equality cases; then exact sufficient conditions for an N -state ordering.
2. **Symbolic checks:** verify stationary identities, derivatives, and pairwise differences.
3. **Numerics:** parameter grids around $a^\dagger$ and b ; separate evidence from proof.
4. **Lean:** formalize
 - ▶ the two-state stationary share,
 - ▶ the derivative-sign equivalence,
 - ▶ the pairwise condition, including positivity and piecewise cases.

Proof statement first; simulations and Lean only after it is fixed.

- ▶ **Different N -state thresholds.** State sufficient rate-by-rate conditions; retain the exact two-state result.
- ▶ **A local result is read as global.** Report both thresholds $a^\dagger$ and b , with every domain restriction.
- ▶ **Myopia omits the value of future skill.** Keep it for baseline comparability; state forward-looking choice as a limitation.
- ▶ $\lambda(e, a)$ **is not jointly concave.** Require concavity in effort conditional on a , which is all the result uses.

Question for discussion: is local reversal the right scope?